Does Hybridization Help? A Leakage-Free, Subject-Level Benchmark of 3D CNN, Transformer, and Hybrid Models for Alzheimer's Disease Classification from Structural MRI

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***Abstract*—Many deep-learning studies of Alzheimer's disease (AD) report near-perfect accuracy from structural MRI, but they often split data at the level of 2D slices, so slices of the same patient appear in both training and test sets. We present a leakage-free benchmark of fully three-dimensional models on 1,132 T1-weighted MRI volumes from 395 subjects of the Alzheimer's Disease Neuroimaging Initiative (ADNI). All models share one preprocessing pipeline, one classifier head, and identical subject-level five-fold splits. We compare a 3D DenseNet-121, a MedicalNet-pretrained 3D ResNet-18, a 3D Vision Transformer (ViT), a Swin Transformer encoder, and two late-fusion hybrids (CNN–ViT and Swin–ViT) on cognitively normal (CN) vs. AD and CN vs. mild cognitive impairment (MCI) vs. AD classification. The CNN–ViT hybrid reaches the highest binary accuracy ($82.2 \pm 5.7\%$, macro-F1 0.80) but is statistically indistinguishable from the from-scratch DenseNet (81.0%; paired $p \approx 0.54$); the hybrid without a convolutional branch is far weaker (70.2%). In the three-class task the best model (Swin, 57.7%) is only about five points above the majority-class baseline, and MCI is the main source of error. A from-scratch ViT fails to learn. A 3D attribution analysis shows that, without skull-stripping, explanation methods disagree and partly highlight non-brain tissue. The results indicate that under honest evaluation a well-trained 3D CNN is a strong baseline that transformer hybrids do not measurably improve at this data scale.**

***Index Terms*—Alzheimer's disease, structural MRI, 3D convolutional neural network, vision transformer, data leakage, explainable AI**

I. Introduction

Alzheimer's disease (AD) is the most common cause of dementia, and structural MRI captures its characteristic atrophy of the hippocampus and medial temporal lobe and the resulting ventricular enlargement [1], [2]. Deep learning has been applied widely to MRI-based AD classification, and recent work increasingly combines convolutional neural networks (CNNs), which capture local texture, with vision transformers (ViTs), which model long-range context [3].

Two problems limit what this literature tells us. First, many studies classify 2D slices and split the data at slice level, so slices of the same patient fall on both sides of the train/test boundary. Yagis *et al.* [4] showed that such leakage raises accuracy from about 66–70% to 95–97% on the same data, and Wen *et al.* [5] found signs of leakage in more than half of the studies they reviewed. Second, hybrid CNN–transformer models are rarely compared against equally tuned single-model baselines under the same splits, so it is unclear whether fusion itself helps.

This paper addresses both problems with a controlled benchmark. Its contributions are:

- A subject-level, five-fold evaluation of six fully 3D models on one curated ADNI cohort, with a shared pipeline, classifier head, and splits, and with majority-class baselines and paired statistical tests reported alongside every comparison.
- Evidence that a CNN–ViT hybrid does not measurably outperform a strong from-scratch 3D CNN, whereas removing the CNN branch costs about 12 accuracy points.
- Two documented negative results: instability of a medically pretrained ResNet, and failure of a from-scratch 3D ViT in the three-class task.
- A 3D explainability analysis that reports where attribution methods disagree, rather than only showing favourable maps.

II. Related Work

Slice-based studies. ViT and CNN models trained on 2D slices report 97–99.8% accuracy for AD staging [6]–[9], and a CNN–Swin hybrid reaches 95.3% on OASIS slices [10]. These pipelines either split data at slice level or use curated 2D Kaggle sets, so their figures are not comparable with subject-level results [4].

Subject-level 3D studies. Under subject-level evaluation, reported performance is markedly lower. Wen *et al.* [5] obtained 85–88% for AD vs. CN with 3D CNNs in a reproducible framework, and Bäckström *et al.* [11] showed that subject-level splitting is essential for 3D CNNs. Saoud and AlMarzouqi [12] used one baseline scan per subject and an ensemble of 138 region-specific 3D ViTs, reporting 90% for AD vs. CN but only 60% for the three-class task. Our work follows this subject-level line and focuses on whether hybridization adds value over strong single models.

TABLE I
CURATED ADNI COHORT

Group	Subjects	Scans	Task
CN	206	627	binary, 3-class
MCI	80	182	3-class
AD	109	323	binary, 3-class
Total	395	1,132	

III. MATERIALS AND METHODS

A. Data

We used ADNI [13], [14] 3T T1-weighted MPRAGE volumes (ADNI GO/2/3) in the pre-processed N3m NIfTI form. We removed 18 scans from 9 subjects that were hippocampal segmentations or followed an off-pipeline protocol; all 9 subjects kept their standard scans. Table I summarizes the cohort. Subjects contribute 2.9 scans on average, so a subject-level split is essential.

B. Evaluation Protocol

We built stratified five-fold splits at the *subject* level (seed 42), so that no subject's scans appear in both training and test data; this is asserted in code. The same splits are used for every model. **[TODO: state where the validation subjects used for early stopping come from in each fold, e.g. an inner split of the four training folds]**. Scan-level outputs are averaged per subject to give one subject-level prediction, and all metrics are computed at subject level. Because the classes are imbalanced, we use macro-F1 as the primary metric and report accuracy and weighted-F1 as secondary metrics. Always predicting CN gives the majority-class baseline: 65.4% accuracy in the binary task and 52.2% in the three-class task. We report mean $\pm$ population standard deviation over folds, and compare models with a paired t-test on fold-wise macro-F1.

C. Preprocessing

A deterministic MONAI [15] pipeline reorients each volume to RAS, resamples it to 1.5 mm isotropic spacing, crops the foreground, applies z-score normalization over non-zero voxels, and resizes it to 96^3 (all binary runs and the three-class ViT) or 128^3 (three-class CNN and Swin). Volumes are not skull-stripped or registered. Training-only augmentation includes random flips, affine transforms ($\pm 10°$), Gaussian noise and smoothing, bias field, gamma, and intensity scaling.

D. Models

All backbones expose a feature vector that feeds a shared dropout (0.2) and linear classifier head. (i) **DenseNet-121** [16]: 3D, trained from scratch, 1024-D features. (ii) **ResNet-18** initialized with MedicalNet weights [17], 512-D. (iii) **ViT** [18]: hidden size 384, 6 layers, 6 heads, 16^3 patches (216 tokens), trained from scratch. (iv) **Swin**: the SwinUNETR encoder [19], [20] (feature size 48, four stages) with global average pooling. (v) **CNN–ViT hybrid** (Fig. 1): DenseNet and ViT branches process the same volume, and their features are concatenated (1408-D) before the head. (vi) **Swin–ViT hybrid**: the same design with Swin in place of DenseNet, which acts as a control for the contribution of the CNN branch.

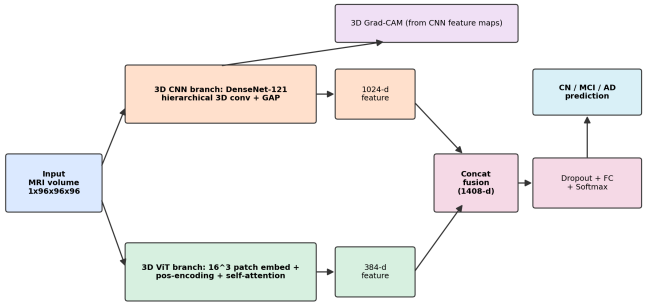


Fig. 1. CNN–ViT hybrid. A 3D DenseNet-121 branch (local features, 1024-D) and a 3D ViT branch (global features, 384-D) process the same volume; their features are concatenated and classified by a shared dropout and linear head.

E. Training

All models use AdamW [21] (learning rate 10^{-4}) with cosine decay and linear warmup, mixed precision, batch size 2 with 4-step gradient accumulation, and gradient clipping at 1.0. Binary runs use class-weighted cross-entropy with label smoothing 0.1 [22], weight decay 0.03, 8 warmup epochs, at most 100 epochs, and early-stopping patience 18. Three-class runs use focal loss [23] ($\gamma = 2$), weight decay 0.01, 5 warmup epochs, at most 80 epochs, and patience 15. **[TODO: state exactly which imbalance corrections were active per run (class weights, weighted sampler, focal loss)]**. Early stopping and checkpoint selection use validation subject-level macro-F1. Training used PyTorch [24] and MONAI on one NVIDIA RTX 5060 Ti (16 GB).

F. Explainability

For trained models we computed Grad-CAM [25], Grad-CAM++ [26], Score-CAM, Integrated Gradients [27], SmoothGrad [28], occlusion, LIME [29], and, for transformers, attention rollout [30], all in 3D.

IV. RESULTS

A. Binary CN vs. AD

Table II reports the binary results. The CNN–ViT hybrid has the highest mean macro-F1 (0.802), followed closely by the from-scratch DenseNet (0.788). Both clearly exceed the majority-class baseline. The Swin–ViT hybrid and the pretrained ResNet are about 12 points lower in accuracy, only 4–5 points above the baseline, and much less stable across folds.

Is the hybrid better? The fold-wise macro-F1 differences between the CNN–ViT hybrid and DenseNet are -0.052, $+0.047$, $+0.065$, -0.013, and $+0.023$ (mean $+0.014$). A paired t-test gives $t = 0.67$ with 4 degrees of freedom ($p \approx 0.54$). The hybrid wins three of five folds and classifies

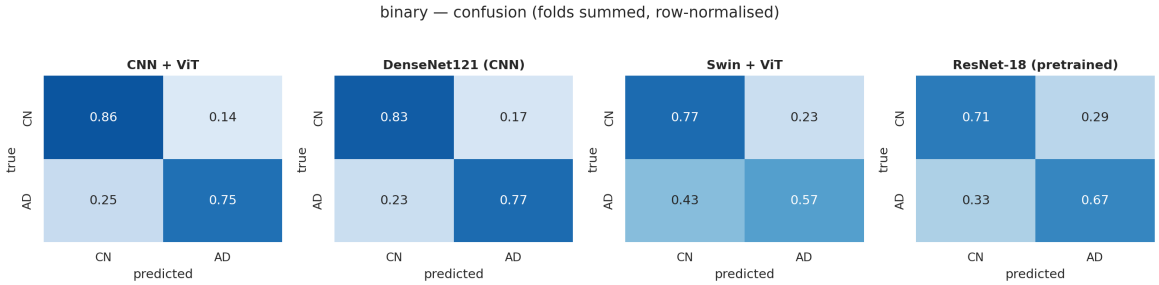


Fig. 2. Binary confusion matrices (folds summed, row-normalized, subject level).

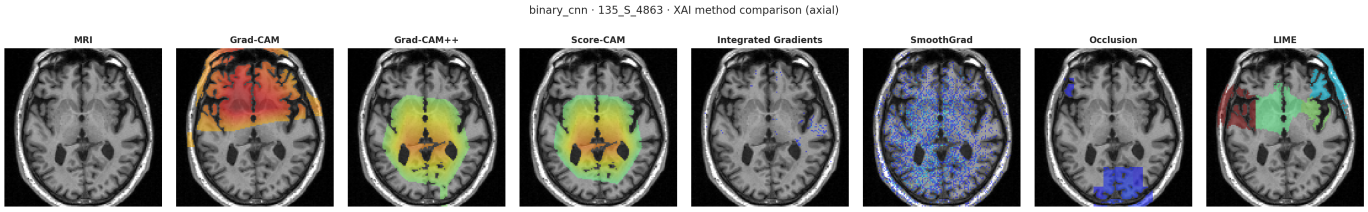


Fig. 3. 3D attribution methods for the binary DenseNet on one AD subject (axial slice). The methods highlight different regions, including non-brain tissue.

TABLE II
BINARY CN VS. AD, SUBJECT-LEVEL 5-FOLD CV (MEAN ± SD)

Model	Accuracy	Macro-F1	W-F1
CNN–ViT hybrid	**0.822**±0.057	**0.802**±0.062	0.820
DenseNet-121 (scratch)	0.810±0.041	0.788±0.062	0.807
Swin–ViT hybrid	0.702±0.070	0.662±0.101	0.696
ResNet-18 (MedicalNet)	0.698±0.139	0.654±0.168	0.678
Majority baseline	*0.654*	*0.395*	*0.517*

TABLE III
THREE-CLASS CN/MCI/AD, SUBJECT-LEVEL 5-FOLD CV

Model	Accuracy	Macro-F1	W-F1
Swin	**0.577**±0.098	**0.537**±0.078	0.573
DenseNet-121	0.572±0.092	0.524±0.071	0.560
ViT (scratch)	0.286±0.015	0.256±0.012	0.184
Majority baseline	*0.522*	*0.229*	*0.358*

259 of 315 subjects correctly against DenseNet's 255. We therefore find no evidence that adding the ViT branch helps. Removing the CNN branch, however, is costly: Swin–ViT loses about 14 points of macro-F1, which indicates that the convolutional branch carries most of the signal at this data scale.

Fig. 2 shows the confusion matrices. CNN–ViT and DenseNet recover both classes with balanced errors (CN recall 0.83–0.86, AD recall 0.75–0.77), whereas Swin–ViT misses 43% of AD subjects.

B. Three-Class CN vs. MCI vs. AD

Table III reports the three-class results. Swin and DenseNet are tied ($t = 0.70$, $p \approx 0.5$) at about 57% accuracy. This is only five points above the majority baseline, although their macro-F1 is more than twice the baseline's. The pooled per-class F1 is about 0.65 for CN, 0.55–0.61 for AD, and only 0.37–0.41 for MCI; most MCI subjects are assigned to CN or AD. For DenseNet, the one-vs-rest AUCs are 0.69 (CN), 0.66 (MCI), and 0.77 (AD). The CNN–ViT hybrid was not evaluated on this task.

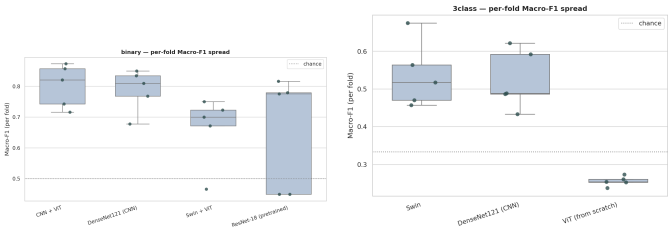


Fig. 4. Per-fold macro-F1: binary (left) and three-class (right).

C. Fold Variance and Negative Results

Fig. 4 shows the fold-wise spread. The pretrained ResNet is bimodal: three folds reach 0.78–0.82 macro-F1 while two fall to 0.449. One possible explanation is that a late accuracy take-off interacts with the shared early-stopping patience, but we have not tested this. The from-scratch ViT falls *below* the majority baseline in the three-class task: it never predicts CN and splits subjects between MCI and AD. This is consistent with the data demands of a ViT trained from scratch [18], possibly compounded by imbalance corrections that push a model that has learned nothing useful toward minority classes.

D. Explainability

Fig. 3 compares the attribution methods for the binary DenseNet on one AD subject. The methods do not agree: Grad-CAM highlights frontal regions, Grad-CAM++ and Score-CAM a central peri-ventricular area, occlusion the occipital region and posterior skull, and LIME partly the scalp. Integrated Gradients is sparse on this slice. For the Swin and Swin–ViT models, group-averaged Integrated Gradients over 16 AD subjects is close to uniform across spatial thirds of the volume, and several maps fall on extra-cerebral tissue. Because the volumes are not skull-stripped, these maps cannot show that the models rely on AD-specific anatomy.

V. Discussion

Under subject-level evaluation our binary result (82%) lies in the same regime as other subject-level 3D studies, slightly below the 85–88% of Wen *et al.* [5] and the 90% of the 138-model ensemble of [12]. Our three-class result (58%) is close to the 60% reported by [12]. The remaining gap is plausibly explained by our minimal preprocessing (no skull-stripping or atlas registration) and reduced 96^3 input resolution. The 97–99% reported by slice-level studies are not comparable.

The main finding is negative but useful. At a cohort of a few hundred subjects, a well-tuned 3D CNN is a strong baseline, and late fusion with a ViT gives no measurable gain, while transformer-only configurations are clearly weaker. Claims of hybrid superiority should therefore be supported by matched single-model baselines and paired tests on identical subject-level splits.

Limitations.

- *Incomplete model grid.* Standalone ViT and Swin were not run on the binary task, and CNN–ViT was not run on the three-class task.
- *Statistical power.* Results come from a single seed and five folds, so differences of one or two points are within noise.
- *Possible age confound.* AD subjects are older on average than CN subjects, and we did not run an age-only baseline.
- *Preprocessing and explainability.* Volumes were not skull-stripped, so the explainability analysis is qualitative.
- *External validation.* The study uses only the ADNI cohort.

VI. Conclusion

We benchmarked six 3D models for AD classification from structural MRI under a strictly subject-level protocol. The CNN–ViT hybrid achieved the highest binary accuracy (82.2%), but it was statistically tied with a from-scratch 3D DenseNet. Three-class performance was limited by MCI, and transformer-only and medically pretrained models were weaker or unstable. Future work will complete the model grid with repeated cross-validation, add skull-stripping and atlas-based ROI attribution, and validate the models on independent cohorts.

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References

[1] P. Scheltens, K. Blennow, M. M. Breteler, B. de Strooper, G. B. Frisoni, S. Salloway, and W. M. Van der Flier, "Alzheimer's disease," *The Lancet*, vol. 388, no. 10043, pp. 505–517, 2016.

[2] G. B. Frisoni, N. C. Fox, C. R. Jack Jr, P. Scheltens, and P. M. Thompson, "The clinical use of structural MRI in alzheimer disease," *Nature Reviews Neurology*, vol. 6, no. 2, pp. 67–77, 2010.

[3] M. A. Bravo-Ortiz, S. A. Holguin-Garcia, S. Quiñones-Arredondo, A. Mora-Rubio, E. Guevara-Navarro, H. B. Arteaga-Arteaga, G. A. Ruz, and R. Tabares-Soto, "A systematic review of vision transformers and convolutional neural networks for alzheimer's disease classification using 3d mri images," *Neural Computing and Applications*, vol. 36, no. 35, pp. 21 985–22 012, 2024.

[4] E. Yagis, S. W. Atnafu, A. García Seco de Herrera, C. Marzi, R. Scheda, M. Giannelli, C. Tessa, L. Citi, and S. Diciotti, "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," *Scientific Reports*, vol. 11, no. 1, p. 22544, 2021.

[5] J. Wen, E. Thibeau-Sutre, M. Diaz-Melo, J. Samper-González, A. Routier, S. Bottani, D. Dormont, S. Durrleman, N. Burgos, and O. Colliot, "Convolutional neural networks for classification of alzheimer's disease: Overview and reproducible evaluation," *Medical Image Analysis*, vol. 63, p. 101694, 2020.

[6] M. H. Alshayeji, "Alzheimer's disease detection and stage identification from magnetic resonance brain images using vision transformer," *Machine Learning: Science and Technology*, vol. 5, no. 3, p. 035011, 2024.

[7] S. Mahim, M. S. Ali, M. O. Hasan, A. A. N. Nafi, A. Sadat, S. Al Hasan, B. Shareef, M. M. Ahsan, M. K. Islam, M. S. Miah *et al.*, "Unlocking the potential of xai for improved alzheimer's disease detection and classification using a vit-gru model," *IEEE Access*, vol. 12, pp. 8390–8412, 2024.

[8] A. El-Assy, H. M. Amer, H. Ibrahim, and M. Mohamed, "A novel cnn architecture for accurate early detection and classification of alzheimer's disease using mri data," *Scientific Reports*, vol. 14, no. 1, p. 3463, 2024.

[9] D. AlSaeed and S. F. Omar, "Brain mri analysis for alzheimer's disease diagnosis using cnn-based feature extraction and machine learning," *Sensors*, vol. 22, no. 8, p. 2911, 2022.

[10] K. Velu and N. Jaisankar, "Design of a cnn–swin transformer model for alzheimer's disease prediction using mri images," *IEEE Access*, 2025.

[11] K. Bäckström, M. Nazari, I. Y.-H. Gu, and A. S. Jakola, "An efficient 3D deep convolutional network for alzheimer's disease diagnosis using MR images," in *IEEE 15th International Symposium on Biomedical Imaging (ISBI)*. IEEE, 2018, pp. 149–153.

[12] L. S. Saoud and H. AlMarzouqi, "Explainable early detection of alzheimer's disease using rois and an ensemble of 138 3d vision transformers," *Scientific Reports*, vol. 14, no. 1, p. 27756, 2024.

[13] C. R. Jack Jr, M. A. Bernstein, N. C. Fox, P. Thompson, G. Alexander, D. Harvey, B. Borowski, P. J. Britson, J. L Whitwell, C. Ward *et al.*, "The alzheimer's disease neuroimaging initiative (ADNI): MRI methods," *Journal of Magnetic Resonance Imaging*, vol. 27, no. 4, pp. 685–691, 2008.

[14] R. C. Petersen, P. S. Aisen, L. A. Beckett, M. C. Donohue, A. C. Gamst, D. J. Harvey, C. R. Jack Jr, W. J. Jagust, L. M. Shaw, A. W. Toga, J. Q. Trojanowski, and M. W. Weiner, "Alzheimer's disease neuroimaging initiative (ADNI): clinical characterization," *Neurology*, vol. 74, no. 3, pp. 201–209, 2010.

[15] M. J. Cardoso, W. Li, R. Brown, N. Ma, E. Kerfoot, Y. Wang, B. Murrey *et al.*, "MONAI: An open-source framework for deep learning in healthcare," 2022.

[16] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.

[17] S. Chen, K. Ma, and Y. Zheng, "Med3D: Transfer learning for 3D medical image analysis," 2019.

[18] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[19] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *International MICCAI Brainlesion Workshop*. Springer, 2021, pp. 272–284.

[20] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10 012–10 022.

[21] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[22] R. Müller, S. Kornblith, and G. E. Hinton, "When does label smoothing help?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[23] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[24] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 8026–8037.

[25] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[26] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847.

[27] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 3319–3328.

[28] D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, "SmoothGrad: removing noise by adding noise," 2017.

[29] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should I trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 1135–1144.

[30] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197.